

Suggested solutions

- 1 a) (i) Explain why the temperature rise is different in the two experiments.
For experiment 1, no work is done by gas, hence all the thermal energy supplied is used to increase the internal energy while for experiment 2, some of the thermal energy supplied is used by the gas to do work in expanding, [2]
resulting in less increase in internal energy [B1]

Hence experiment 2 has a smaller increase in internal energy than experiment 1, resulting in smaller temperature rise [B1]

(Alternatively, use first law of thermodynamics to explain)

(Important: Comparisons between both experiments are required)

(ii)

	thermal energy supplied to gas/J	work done on gas /J	increase in internal energy of gas/J
Experiment 1	+150	0 [1 mark]	+150 [1 mark]
Experiment 2	+150	-57 [1 mark]	+ 93 [1 mark]

- 1 mark for calculating increase in internal energy for experiment 2, with working
 $\Delta U_2 = \Delta U_1 (\Delta T_2 / \Delta T_1) = 150 (18 / 29) = 93 \text{ J}$

- 1 mark for calculating work done on gas in experiment 2, with working
 $W = \Delta U - Q = 93 - 150 = - 57 \text{ J}$

$$\begin{aligned}
 \text{(b) (i)} \quad KE &= \frac{3}{2} kT && \text{[M1]} \\
 &= \frac{3}{2} (1.38 \times 10^{-23})(100 + 273) \\
 &= 7.72 \times 10^{-21} \text{ J} && \text{[A1]}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) (ii)} \quad m_{\text{Ar}} \langle c^2 \rangle_{\text{Ar}} &= 7.72 \times 10^{-21} \\
 0.5(1.8)(3.35 \times 10^{-26}) \langle c^2 \rangle_{\text{Ar}} &= 7.72 \times 10^{-21} && \text{[M1]} \\
 V_{\text{rms,Ar}} &= 506 \text{ m s}^{-1} && \text{[A1]}
 \end{aligned}$$